

IIT Placement Intelligence Platform

System Architecture & Project Blueprint

Problem Statement

Placement information across IITs is scattered across reports, brochures and news sources. This platform centralizes placement statistics and provides analytics, comparisons and predictive insights.

Technology Stack

Frontend: React, Tailwind CSS, Recharts

Backend: FastAPI (Python)

Database: MySQL

Machine Learning: Pandas, NumPy, Scikit-Learn, Random Forest Regressor

Deployment: Vercel + Render + MySQL Cloud

Version Control: Git & GitHub

System Architecture

React Frontend → FastAPI Backend → MySQL Database & ML Service → JSON Responses → Dashboard

Frontend Structure

Pages: Dashboard, IIT Comparison, Branch Analytics, Sector Analytics, Predictions

Components: Navbar, Filters, Stats Cards, Insight Cards, Charts

Backend Structure

Routes: Dashboard, Comparison, Analytics, Prediction

Services: Analytics Service, Prediction Service

Database Layer: Connection, Models

ML Layer: Training and Prediction Modules

Database Design

`institutes(id, name)`

`placement_stats(id, institute_id, year, program, branch, avg_package, median_package, highest_domestic, highest_international, placement_percentage)`

`sector_stats(id, institute_id, year, branch, sector, students_placed, placement_percentage)`

Core Features

1. Dashboard
2. IIT Comparison
3. Branch Analytics
4. Sector Analytics
5. Placement Prediction

Machine Learning

Model 1: Average Package Prediction

Model 2: Placement Percentage Prediction

Algorithm: Random Forest Regressor

Metrics: MAE, RMSE, R^2

Deployment Architecture

Frontend deployed on Vercel, Backend deployed on Render, MySQL hosted on a managed cloud database.